

Tharmapalan Thanujan

Machine Learning Engineer / Software Engineer

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Machine Learning Engineer building scalable, explainable AI systems for real-world impact. Boosted model performance in production, deployed cloud inference pipelines, and led data-centric ML initiatives. Passionate about responsible, high-impact AI at scale

Education

University Of Peradeniya

BSc.Eng(Hons.) In Computer Engineering

GPA: 3.5/4.0

Nov.2018 - Dec.2023

Professional Experience

Senior Machine Learning Engineer

Jun 2026 - Present

Veracity Digital

- Lead end-to-end development of computer vision initiatives, owning solution architecture, data strategy, model development, deployment, and production integration.
- Design AI/ML architectures and technical proposals for client and pre-sales engagements, translating business requirements into scalable implementation roadmaps.
- Leading development of an AI-powered geospatial intelligence platform using satellite and multispectral imagery for plantation monitoring and decision support.
- **Technologies:** Python, PyTorch, OpenCV, YOLO, Flask, Docker, GCP, AWS, PostgreSQL, FiftyOne, DVC, Git.

Machine Learning Engineer

Jan 2024 - Jun 2026

Spectrify AI

- Owned end-to-end development of Spectrify AI's tea leaf quality grading system, delivering dataset creation, annotation QA, model training, cloud inference deployment, and Flutter-based field integration.
- Improved type-leaf classification accuracy from 80% to 90% and achieved over 85% multi-class tea leaf grading accuracy under real-world field conditions.
- Integrated ML inference with a Flutter mobile application and backend platform, enabling authenticated scan capture, cloud-based prediction, and real-time result visualization.
- Designed and deployed an asynchronous GPU-based Flask inference API on GCP using Docker, reducing infrastructure cost by 10× while supporting scalable real-time inference.
- Led five annotators and established annotation QA workflows, improving dataset quality by 10% and reducing labeling costs by 6×.
- Developed a 3D floor-plan generation system from architectural layouts, achieving approximately 90% accuracy in automated spatial visualization.
- Prepared BRDs, technical proposals, delivery plans, and ML-to-business roadmaps to align product development with customer requirements and business goals.
- **Technologies:** Python, PyTorch, Flask, Docker, GCP, AWS, Flutter, NestJS, PostgreSQL, NATS, FiftyOne, DVC, Git.

Associate Machine Learning Engineer / Consultant

June 2023 - Dec 2023

SpectrifyAI

- Integrated an Active Learning pipeline to prioritize uncertain samples, reducing annotation cost and improving labeling efficiency.
- Developed and fine-tuned object detection and segmentation models using sliding-window inference for high-resolution image analysis.
- Improved mAP50 from 29% to 80% through model tuning, error analysis, and systematic evaluation of performance bottlenecks.
- Designed and prototyped a Raspberry Pi-based vision device for real-time object detection applications.
- **Technologies:** Python, PyTorch, AWS, Git, Raspberry Pi.

Machine Learning Engineer Intern

Dec 2022 - June 2023

Veracity AI and Spectrify AI

- Benchmarked object detection models based on accuracy, inference speed, and memory efficiency for deployment-ready computer vision applications.
- Built and maintained data pipelines for visualization, de-duplication, augmentation, and dataset quality control.
- Improved annotation workflows by defining labeling guidelines, selecting tools, and streamlining team processes to reduce labeling cost and rework.
- Trained and optimized object detection and face recognition models using hyperparameter tuning, Grid Search, Random Search, and Bayesian Optimization.
- **Technologies:** Python, PyTorch, AWS, Git

Projects

Studying the Barriers to Using Explainable ML to Improve Human-ML Collaboration | Group |

2023 - Present

- Research project: Validating Explainable ML tools to ensure transparency and interpretability in complex models and mitigate potential disagreements within Explainable AI models.
- Contribution :
 - Developed an end-to-end Explainable ML pipeline integrating multiple interpretability methods (e.g., SHAP, LIME, Grad-CAM) for tabular and high-stake decision models.
 - Designed and implemented algorithms to quantify disagreement among explanation metrics and analyzed model behavior across local and global interpretability settings.
 - Conducted exploratory data analysis (EDA), feature compacting, and model sanity checks to ensure robust and reproducible results.
 - Refactored the pipeline into modular components following software engineering best practices (OOP, SOLID).
- **Technologies:** Python, Scikit-learn, Explainable AI (SHAP, LIME), Machine Learning, Algorithm Design, Data Analysis.

Leaf disease classification | Group |

2020 - 2022

- Research project on leaf disease classification using deep learning and explainable AI techniques.
- Contributions:
 - Developed a web application to classify tomato plant diseases based on leaf images.
 - Collected, cleaned, and augmented image datasets; trained an image classification model using MobileNet and transfer learning.
 - Evaluated model robustness by analyzing performance under various noise perturbations.
 - Applied explainable AI methods such as Vanilla Saliency Maps, Grad-CAM, Grad-CAM++, SHAP, and LIME to interpret model predictions.
 - Integrated the trained model into a Django web application for real-time disease prediction and visualization.
- **Technologies:** Python, TensorFlow, Keras, Django, Transfer Learning, Explainable AI, Image Processing.

Host Pathogen Interaction | Group |

2021 - 2022

- Developed a Django-based web platform for microbiome DNA analysis and disease prediction using machine learning techniques.
- Contributions :
 - Implemented data preprocessing pipelines and collaborated on integrating a TensorFlow-based classification model for disease inference.
 - Containerized the application using Docker and configured Gunicorn for scalable, production-grade deployment.
 - Adopted Agile development methodology to manage project tasks and iterative improvements.
- **Technologies:** Django, Machine Learning, DevOps

Smart Inventory Management System | Group |

2021 - 2022

- Developed a web application using Laravel and Mysql to manage inventory items in the computer engineering department.
- Contributions :
 - Designed and implemented the database for the Inventory Management System.
 - Developed a back-end for the web application.
- **Technologies:** Laravel, MySQL

Certifications

Deep Learning Specialization |

Neural Networks, CNNs, Sequence Models – Applied in production-level ML pipelines

Generative Adversarial Networks (GANs) |

Implemented GANs for image synthesis and data augmentation projects

Server-side Dev & Full-stack Web |

NodeJS, Express, MongoDB, React – Production-ready web applications

Algorithms & Data Structures |

problem-solving skills, Efficient coding solutions

Skills

Machine Learning / AI

TensorFlow, PyTorch, Keras, Scikit-learn, Hugging Face, OpenAI API, Gemini API, Vision-Language Models (VLMs), Object Detection, Image Segmentation, Explainable AI (SHAP, LIME, Grad-CAM), Model Optimization

MLOps / DevOps

GCP, AWS, MLflow, WandB, DVC, Docker, CI/CD pipelines, GPU Inference Pipelines

Programming / System Design

Python, Java, C++, SQL, Verilog, Embedded Systems

Web / Backend / Frontend

NodeJS, Django, Laravel, ReactJS, SpringBoot, NestJS, HTML/CSS, Redux, REST APIs

References

Prof. Roshan G. Ragel

Former Head of Department

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Department of Computer Engineering

University of Peradeniya

Keerththanan P.

Associate Tech Lead

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Aventage